Functions, Modulus and Quadratics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Functions, Modulus & Quadratics Practice Exercises**

[Functions, Modulus & Quadratic Theory · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

2 **[Foundational]** • **Target: Functions, Modulus & Quadratics Practice Exercises**

[Differentiation, Rules & Rates of Change · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

3 **[Foundational]** • **Target: Functions, Modulus & Quadratics Practice Exercises**

[Integration, Areas & Kinematics · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

4 **[Foundational]** • **Target: Functions, Modulus & Quadratics Practice Exercises**

[Logarithms, Exponentials & Straight-Line Reduction · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

5 **[Foundational]** • **Target: Functions, Modulus & Quadratics Practice Exercises**

[Permutations & Combinations · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

6 **[Foundational]** • **Target: Functions, Modulus & Quadratics Practice Exercises**

[Polynomials, Remainder Theorem & Binomial Theorem · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

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7 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

[Trigonometry & Circular Measure · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Vectors in Two Dimensions & Relative Motion · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

9 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Functions: Modulus & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Indices, Surds & Quadratics · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

11 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Functions: Composite & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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14 **[Intermediate]** • **Target: Functions, Modulus & Quadratics Practice Exercises**

Given $f(x) = 2x - 3$ for $x \in \mathbb{R}$, sketch $y = |f(x)|$ and solve $|2x - 3| = 5$.

Sketch: V-shape with vertex at $(1.5, 0)$. Solve: $2x - 3 = 5 \Rightarrow x = 4$ or $2x - 3 = -5 \Rightarrow x = -1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 2.01 Simultaneous Equations and Inequalities**

Solve the quadratic equation $2x^2 - 5x - 3 = 0$ by completing the square. State the vertex of the corresponding parabola $y = 2x^2 - 5x - 3$.

Factor 2 from the first two terms: $2\left(x^2 - \frac{5}{2}x\right) - 3 = 0$. Halve the coefficient of x: $\frac{1}{2}\left(-\frac{5}{2}\right) = -\frac{5}{4}$. Complete the square: $2\left(x - \frac{5}{4}\right)^2 - 2\left(\frac{25}{16}\right) - 3 = 0 \Rightarrow 2\left(x - \frac{5}{4}\right)^2 - \frac{25}{8} - \frac{24}{8} = 0 \Rightarrow 2\left(x - \frac{5}{4}\right)^2 = \frac{49}{8}$. Divide by 2: $\left(x - \frac{5}{4}\right)^2 = \frac{49}{16}$. Take square roots.

16 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 3.01 Indices, Surds and Polynomials**

Find the set of values of the constant k for which the equation $kx^2 + 4x + (k - 3) = 0$ has real roots.

For real roots, the discriminant $\Delta = b^2 - 4ac \geq 0$. Here $a = k$, $b = 4$, $c = k - 3$:
$\Delta = 4^2 - 4(k)(k - 3) = 16 - 4k^2 + 12k \geq 0$ Divide the entire inequality by -4, reversing the inequality sign: $k^2 - 3k - 4 \leq 0$ Factor the quadratic: $(k - 4)(k + 1) \leq 0$. The critical values are $k = -1$ and $k = 4$. Since the parabola opens upward, the product is ≤ 0 between the roots: $-1 \leq k \leq 4$. Note that if $k = 0$, the equation becomes the linear equation $4x - 3 = 0$, which has one r.

17 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 4.01 Logarithmic and Exponential Functions**

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

Since $\deg(\text{num}) = 3$ and $\deg(\text{denom}) = 2$ with $3 - 2 = 1$, a slant asymptote exists. Perform polynomial long division: $(2x^3 - 3x^2 + 5) \div (x^2 + 1) = (2x - 3) + \frac{-2x + 8}{x^2 + 1}$. The linear quotient gives the slant asymptote: $y = 2x - 3$. To find where the graph intersects this asymptote, set the remainder equal to zero: $\frac{-2x + 8}{x^2 + 1} = 0 \Rightarrow -2x + 8 = 0 \Rightarrow x = 4$. Substitute $x = 4$ into the asymptote equation: $y = 2(4) - 3 = 5$. The curve intersects.

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18 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure**

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

Factor numerator and denominator completely: $R(x) = \frac{(2x-1)(x+3)}{(x+1)(x+3)}$. The factor $(x+3)$ cancels for $x \neq -3$. Evaluate the simplified form $g(x) = \frac{2x-1}{x+1}$ at $x = -3$: $g(-3) = \frac{2(-3)-1}{-3+1} = \frac{-7}{-2} = 3.5$. Key features:Hole: $(-3, 3.5)$Vertical asymptote: $x + 1 = 0 \Rightarrow x = -1$Horizontal asymptote: Equal degrees (2 and 2) $\Rightarrow y = 2/1 = 2$$x$-intercept: Numerator root $2x - 1 = 0 \Rightarrow (0.5, 0)$$y$-intercept.

19 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 6.01 Permutations and Combinations**

The function f is defined by $f(x) = x^2 - 6x + 13$ for $x \geq 3$. (a) State the range of f. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1}.

(a) Complete the square: $f(x) = (x-3)^2 - 9 + 13 = (x-3)^2 + 4$. Since $x \geq 3$, $(x-3)^2 \geq 0$, so the minimum value is $f(3) = 4$. The range of f is $f(x) \geq 4$. (b) Set $y = (x-3)^2 + 4$. Rearrange for x: $y - 4 = (x-3)^2 \Rightarrow x - 3 = +\sqrt{y-4}$ (since $x \geq 3$) $x = 3 + \sqrt{y-4}$ Therefore, $f^{-1}(x) = 3 + \sqrt{x-4}$. Domain of f^{-1} is $x \geq 4$ (the range of f), and range of f^{-1} is $f^{-1}(x) \geq 3$ (the domain of f).

20 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 7.01 Differentiation and Rates of Change**

Functions g and h are defined by $g(x) = 2x - 1$ for $x \in \mathbb{R}$, and $h(x) = \frac{6}{x-2}$ for $x \neq 2$. (a) Find an expression for $hg(x)$ in simplest form. (b) Solve the equation $hg(x) = g(x)$.

(a) Substitute $g(x)$ into h: $hg(x) = h(2x-1) = \frac{6}{(2x-1)-2} = \frac{6}{2x-3}$ $\left(x \neq \frac{3}{2}\right)$ (b) Set $hg(x) = g(x)$: $\frac{6}{2x-3} = 2x - 1 \Rightarrow 6 = (2x-1)(2x-3)$ Expand the right-hand side: $6 = 4x^2 - 8x + 3 \Rightarrow 4x^2 - 8x - 3 = 0$ Factor: $(2x-3)(2x+1) = 0$. Since $x \neq \frac{3}{2}$ for $hg(x)$ to be defined, the value $x = \frac{3}{2}$ is excluded. Therefore, the unique solution is $2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation \(|2x + 3| = 7 \).

a. \(x = 3 \) and \(x = -3 \)

b. \(x = -2 \) and \(x = -\frac{10}{2} \)

c. \(x = 2 \) only

d. **\(x = 2 \) and \(x = -\frac{10}{2} \)** ✓

Case 1: \(2x + 3 = 7 \implies 2x = 4 \implies x = 2 \). Case 2: \(2x + 3 = -7 \implies 2x = -10 \implies x = -\frac{10}{2} \).

22 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The function $f(x) = \frac{3x+1}{x-2}$ is defined for $xe2$. Find the inverse function $f^{-1}(x)$.

a. $f^{-1}(x) = \frac{3x-2}{x+1}$

b. $f^{-1}(x) = \frac{2x-1}{x-3}$

c. $f^{-1}(x) = \frac{x-2}{3x+1}$

d. $f^{-1}(x) = \frac{2x+1}{x-3} \quad (xe3)$ ✓

Let $y = \frac{3x+1}{x-2} \Rightarrow y(x-2) = 3x+1 \Rightarrow yx - 2y = 3x + 1 \Rightarrow x(y-3) = 2y + 1 \Rightarrow x = \frac{2y+1}{y-3}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-3}$.

23 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Find the set of values of the constant m for which the equation $|2x - 6| = mx + 2$ has exactly two distinct real solutions.

a. $-\frac{2}{3} < m < 2$ ✓

b. $-2 < m < 2$

c. $m2$

d. $-\frac{2}{3} \le m \le 2$

We consider the function $y = |2x - 6|$ and the line $y = mx + 2$. 1. **Analyze the modulus function:** The graph of $y = |2x - 6|$ consists of two linear rays meeting at the vertex $(3, 0)$: \[y = \begin{cases} 2x - 6, & \text{for } x \ge 3 \text{ (gradient } 2\text{)}, \\ -2x + 6, & \text{for } x -\frac{2}{3} \). 4. **Upper boundary (parallelism with the right ray):** The right ray has gradient 2. - If $m = 2$, the line $y = 2x + 2$ is parallel to $y = 2x - 6$ and never intersects the right ray; it intersects.

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24 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Find the set of values of the constant k for which the equation $|x^2 - 4| = 2x + k$ has exactly four distinct real solutions.

a. $4 < k < 5$ ✓

b. $4 \le k \le 5$

c. $-4 < k < 4$

d. $k > 5$

To find the number of solutions to $|x^2 - 4| = 2x + k$, we consider the intersections of the curve $y = |x^2 - 4|$ and the line $y = 2x + k$. The curve is defined piecewise: - Inner arch: $y = 4 - x^2$ for $-2 \le x \le 2$ - Outer branches: $y = x^2 - 4$ for $x2$ 1. **Intersection with the inner arch ($-2 \le x \le 2$):** $4 - x^2 = 2x + k \Leftrightarrow x^2 + 2x + (k - 4) = 0$ For two distinct real intersections, the discriminant must be strictly positive: x The discriminant is.

25 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The functions f and g are defined for all real values of x by $f(x) = 2x - 1$ and $g(x) = \frac{x+5}{3}$. Find the value of x such that $fg(x) = 9$.

a. $x = 10$ ✓

b. $x = \frac{23}{2}$

c. $x = 9$

d. $x = 5$

To find x such that $fg(x) = 9$, we can either determine the composite expression $fg(x)$ or solve sequentially from the outer function to the inner function. **Method 1 (Sequential Evaluation):** We have $f(g(x)) = 9$. Since $f(u) = 2u - 1$, setting $f(u) = 9$ gives: $2u - 1 = 9 \Rightarrow 2u = 10 \Rightarrow u = 5$. Thus, $g(x) = 5$. Now solve for x: $\frac{x+5}{3} = 5 \Rightarrow x + 5 = 15 \Rightarrow x = 10$. **Method 2 (Forming the Composite Function):**.

26 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The quadratic equation $x^2 - (2k + 1)x + (k^2 + 2) = 0$ has real roots α and β. Given that $\alpha^2 + \beta^2 = 27$, find the value of the constant k.

a. 3 ✓

b. -5

c. 3 or -5

d. 5

From the relations between roots and coefficients for the quadratic equation $x^2 - (2k + 1)x + (k^2 + 2) = 0$: $\alpha + \beta = 2k + 1$ and $\alpha\beta = k^2 + 2$ Expressing $\alpha^2 + \beta^2$ in terms of the sum and product of the roots: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ Substitute the known values: $(2k + 1)^2 - 2(k^2 + 2) = 27$ Expand and simplify: $(4k^2 + 4k + 1) - (2k^2 + 4) = 27$ $2k^2 + 4k - 3 = 27 \Rightarrow 2k^2 + 4k - 30 = 0 \Rightarrow k^2 + 2k - 15 = 0$ F.

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27 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The line $y = mx - 1$ intersects the curve $y = x^2 + 3x + k$ at two distinct points for all real values of m. Find the range of values of the constant k.

a. $k < -1$ ✓

b. $k > -1$

c. $k < 1$

d. $k > 1$

To find the points of intersection of the line and the curve, set their equations equal: $x^2 + 3x + k = mx - 1$ Rearranging into standard quadratic form $ax^2 + bx + c = 0$: $x^2 + (3 - m)x + (k + 1) = 0$ For the line and curve to intersect at two distinct points, the discriminant with respect to x, denoted Δ_x, must be strictly positive: $\Delta_x = (3 - m)^2 - 4(1)(k + 1) > 0$ Expanding and simplifying: $\Delta_x = m^2 - 6m + 9 - 4k - 4 = m^2 - 6m + (5 - 4k) > 0$ We are given that this inequality holds.

28 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The function f is defined for $x \in \mathbb{R}$ by $f(x) = x^2 - 2kx + (3k + 10)$, where k is a real constant, and the equation $f(x) = 0$ has no real roots. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - k|$. Find the set of values of k for which the minimum value of the composite function $h(x) = f(g(x))$ is strictly greater than 4.

a. $-2 < k < \frac{3 + \sqrt{33}}{2}$ ✓

b. $\frac{3 - \sqrt{33}}{2} < k < \frac{3 + \sqrt{33}}{2}$

c. $-2 < k < 5$

d. $0 \leq k < \frac{3 + \sqrt{33}}{2}$

We solve the problem in three systematic stages: 1. **Condition from the Discriminant of $f(x) = 0$:** Since $f(x) = 0$ has no real roots, its discriminant Δ must be strictly negative: x Combining with $k < 0$, this yields $-2 < k < 0$. 3. **Combining the Conditions:** Combining Case 1 and Case 2 gives: $(-2, 0) \cup \left[0, \frac{3 + \sqrt{33}}{2}\right) = -2 < k < \frac{3 + \sqrt{33}}{2}$. Since $\frac{3 + \sqrt{33}}{2} \approx 4.372 < 5$, this entire interval lies within the initial constrai.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The function f is defined for $x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Find an expression for $f^{-1}(x)$.

a. $f^{-1}(x) = \frac{3x+1}{x-2}$ ✓

b. $f^{-1}(x) = \frac{x-3}{2x+1}$

c. $f^{-1}(x) = \frac{3x-1}{x+2}$

d. $f^{-1}(x) = \frac{3x+1}{x+2}$

To find the inverse function $f^{-1}(x)$, let $y = f(x)$: $y = \frac{2x+1}{x-3}$ Multiply both sides by $(x-3)$: $y(x-3) = 2x+1$ Expand the left-hand side: $xy - 3y = 2x + 1$ Rearrange to collect all terms involving x on one side: $xy - 2x = 3y + 1$ Factor out x: $x(y-2) = 3y+1$ Divide by $(y-2)$: $x = \frac{3y+1}{y-2}$ Replacing y with x gives: $f^{-1}(x) = \frac{3x+1}{x-2}, \quad x \neq 2$. Common misconceptions: - $\frac{x-3}{2x+1}$ confuses the invers.

30 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The function f is defined by $f(x) = \frac{2x-1}{x+3}, \quad$ for $x \neq -3$. Which of the following expressions represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+1}{2-x}, \quad x \neq 2$ ✓

b. $f^{-1}(x) = \frac{x+3}{2x-1}, \quad x \neq \frac{1}{2}$

c. $f^{-1}(x) = \frac{3x-1}{x-2}, \quad x \neq 2$

d. $f^{-1}(x) = \frac{2x+1}{x-3}, \quad x \neq 3$

To find the inverse function $f^{-1}(x)$: 1. Set $y = f(x)$: $y = \frac{2x-1}{x+3}$ 2. Multiply both sides by the denominator $x+3$: $y(x+3) = 2x - 1 \Rightarrow xy + 3y = 2x - 1$ 3. Collect all terms containing x on one side and constant/y terms on the other: $xy - 2x = -3y - 1$ 4. Factor out x: $x(y-2) = -(3y+1)$ 5. Solve for x: $x = \frac{-(3y+1)}{y-2} = \frac{3y+1}{2-y}$ 6. Express in terms of x: $f^{-1}(x) = \frac{3x+1}{2-x}, \quad x \neq 2$

31 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

A sector of a circle of radius $r = 8$ cm has central angle $\theta = 0.9$ rad. What is the area of the sector?

a. $28.8\ \text{cm}^2$ ✓

b. $57.6\ \text{cm}^2$

c. $33.8\ \text{cm}^2$

d. $7.2\ \text{cm}^2$

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(8)^2(0.9) = 28.8\ \text{cm}^2$.

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32 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the exponential equation $2^{x+2} = 5^x$, giving x in terms of common logarithms.

a. $x = \frac{2\ln 2}{\ln 5 - \ln 2}$ ✓

b. $x = \frac{2\ln 5}{\ln 5 - \ln 2}$

c. $x = \frac{\ln 2}{2(\ln 5 - \ln 2)}$

d. $x = \frac{2}{\ln 5 - \ln 2}$

Take natural logs of both sides: $(x+2)\ln 2 = x\ln 5 \Rightarrow x\ln 2 + 2\ln 2 = x\ln 5 \Rightarrow x(\ln 5 - \ln 2) = 2\ln 2 \Rightarrow x = \frac{2\ln 2}{\ln 5 - \ln 2}$.

33 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x?

a. **12** ✓

b. 3

c. 6

d. 14

From the first equation: $\log_2(x/y) = 2 \Rightarrow x/y = 2^2 = 4 \Rightarrow x = 4y$. Substitute into second: $4y - 3y = 3 \Rightarrow y = 3$. Thus $x = 4(3) = 12$.

34 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

Find all solutions of $2\cos^2 x + 3\sin x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

a. $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$ ✓

b. $x = \frac{\pi}{3},\ \frac{2\pi}{3}$

c. $x = \frac{\pi}{6},\ \frac{7\pi}{6}$

d. $x = 0,\ \pi$

Solving quadratic trig equation $2\cos^2 x + 3\sin x - 3 = 0$ yields solutions $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$.

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35 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The diagram shows part of the curve $y = 9 - x^2$ for $x \geq 0$ and the tangent to the curve at the point $P(2, 5)$. Find the exact area of the finite region bounded by the curve, the tangent at P, and the x-axis.

a. $\frac{11}{24}$ ✓

b. $\frac{125}{192}$

c. $\frac{8}{3}$

d. $\frac{25}{8}$

1. **Find the equation of the tangent at $P(2, 5)$:** Given $y = 9 - x^2$, differentiating with respect to x gives: $\frac{dy}{dx} = -2x$ At $x = 2$, the gradient of the tangent is $m = -2(2) = -4$. The equation of the tangent line is: $y - 5 = -4(x - 2) \Rightarrow y = 13 - 4x$ 2. **Determine the x-intercepts:** - Tangent: Set $y = 0 \Rightarrow 13 - 4x = 0 \Rightarrow x = \frac{13}{4} = 3.25$. - Curve: Set $y = 0 \Rightarrow 9 - x^2 = 0 \Rightarrow x = 3$ (since $x \geq 0$). 3. **Calculate the area of the bou.

36 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The diagram shows part of the curve $y = ax^3 + bx^2 + 9x + 1$, where a and b are constants. The curve has a stationary point at $(1, 5)$. Find the coordinates of the other stationary point on the curve.

a. $(3, 1)$ ✓

b. $(3, 0)$

c. $(3, -1)$

d. $(-3, 1)$

To find the coordinates of the other stationary point, we first determine the unknown parameters a and b. **Step 1: Use the given coordinates of the point on the curve** Since the point $(1, 5)$ lies on the curve: $5 = a(1)^3 + b(1)^2 + 9(1) + 1$ $5 = a + b + 10 \Rightarrow a + b = -5 \quad - - -$ (Equation 1) **Step 2: Differentiate to find $\frac{dy}{dx}$** $\frac{dy}{dx} = 3ax^2 + 2bx + 9$ **Step 3: Apply the stationary condition at $x = 1$** At a stationary poi.

37 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$. Find the area of the region enclosed between the curve and the x-axis.

a. 27

b. **36** ✓

c. 54

d. 72

To find the area enclosed between the curve and the x-axis, integrate y with respect to x from $x = 0$ to $x = 6$: Area $= \int_0^6 (6x - x^2)\,dx$ Integrate term by term: $\int (6x - x^2)\,dx = \left[\frac{6x^2}{2} - \frac{x^3}{3}\right]_0^6 = \left[3x^2 - \frac{x^3}{3}\right]_0^6$ Substitute the limits: $\left(3(6)^2 - \frac{6^3}{3}\right) - 0 = 3(36) - \frac{216}{3} = 108 - 72 = 36$ Common errors: - Estimating the region as a triangle with base 6 and maxim.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.

a. $\frac{32}{3}$ ✓

b. 16

c. $\frac{64}{3}$

d. $\frac{16}{3}$

To find the area enclosed by the curve and the x-axis, first find the x-intercepts by setting $y = 0$: $4x - x^2 = 0 \Rightarrow x(4 - x) = 0 \Rightarrow x = 0$ or $x = 4$. The area is given by the definite integral of y with respect to x from $x = 0$ to $x = 4$: $\int_0^4 (4x - x^2)\,dx = \left[2x^2 - \frac{x^3}{3}\right]_0^4$ Evaluating between the limits: $\left(2(4)^2 - \frac{4^3}{3}\right) - 0 = 32 - \frac{64}{3} = \frac{96 - 64}{3} = \frac{32}{3}$. Common misconceptio.

39 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Find the equation of the tangent line to the curve $y = (2x + 1)^3$ at the point where $x = 0$.

a. $y = 2x + 1$

b. $y = 6x - 1$

c. $y = 6x + 1$ ✓

d. $y = 6x + 3$

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(2x + 1)^2 \cdot (2) = 6(2x + 1)^2$. At $x = 0$, slope $m = 6$. Equation: $y - 1 = 6(x - 0) \Rightarrow y = 6x + 1$.

40 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Two circles, C_1 with centre O_1 and radius 6 cm, and C_2 with centre O_2 and radius $6\sqrt{3}$ cm, intersect at the points A and B. The distance between their centres is $O_1O_2 = 12$ cm. Find the exact area of the region common to both circles, in cm^2.

a. $30\pi - 36\sqrt{3}$ ✓

b. $30\pi + 36\sqrt{3}$

c. $42\pi - 36\sqrt{3}$

d. $15\pi - 18\sqrt{3}$

1. **Determine the angles subtended at each centre:** Consider triangle O_1AO_2 with side lengths $O_1A = 6$, $O_2A = 6\sqrt{3}$, and $O_1O_2 = 12$. Notice that: $(O_1A)^2 + (O_2A)^2 = 6^2 + (6\sqrt{3})^2 = 36 + 108 = 144 = 12^2 = (O_1O_2)^2$ By the converse of Pythagoras' Theorem, $\angle O_1AO_2 = \frac{\pi}{2}$ radians. Using right-angled trigonometry in ΔO_1AO_2.

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41 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The polynomial $P(x)$ is defined by $P(x) = (2 + ax)^n + bx^4 + cx - 4$ where n is an integer greater than 2, and a, b, c are constants with $a > 0$. In the binomial expansion of $(2 + ax)^n$ in ascending powers of x: - the coefficient of x^2 is twice the coefficient of x, - the coefficient of x^3 is equal to the coefficient of x^2. Given that $(x + 2)$ is a factor of $P(x)$, and that the remainder when $P(x)$ is divided by $(x + 1)$ is 7, find the remainder when $P(x)$ is divided by $(x - 1)$.

a. **1011** ✓

b. 1031

c. 1015

d. 499

We solve the problem in three stages. **Stage 1: Determine a and n using the Binomial Theorem** The general term in the expansion of $(2 + ax)^n$ is $\binom{n}{r}2^{n-r}(ax)^r$. - The term in x is $\binom{n}{1}2^{n-1}(ax) = n \cdot 2^{n-1}ax$. - The term in x^2 is $\binom{n}{2}2^{n-2}(ax)^2 = \frac{n(n-1)}{2}2^{n-2}a^2x^2$. - The term in x^3 is $\binom{n}{3}2^{n-3}(ax)^3 = \frac{n(n-1)(n-2)}{6}2^{n-3}a^3x^3$. Using the first condition (coefficient of x^2 is twice that of x).

42 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

At 08:00, a cargo vessel has position vector $(-7\mathbf{i} + 22\mathbf{j})$ km relative to a coastal radar station at the origin O, and moves with a constant velocity of $(15\mathbf{i} + 5\mathbf{j})$ km/h. At the same time (08:00), a high-speed patrol boat stationed at $(5\mathbf{i} + 6\mathbf{j})$ km sets off in a straight line at a constant speed v km/h to intercept the cargo vessel. Find the minimum speed v required for the patrol boat to intercept the cargo vessel, and the time at which interception occurs at this minimum speed.

a. **Minimum speed 15 km/h, interception at 12:00** ✓

b. Minimum speed 15 km/h, interception at 10:00

c. Minimum speed $5\sqrt{10}$ km/h, interception at 12:00

d. Minimum speed 12.5 km/h, interception at 11:12

Let t denote the time in hours elapsed after 08:00 (with $t > 0$). 1. **Position vectors at time t:** - Cargo vessel: $\mathbf{r}_C(t) = (-7\mathbf{i} + 22\mathbf{j}) + t(15\mathbf{i} + 5\mathbf{j}) = (15t - 7)\mathbf{i} + (5t + 22)\mathbf{j}$ - Patrol boat starts at $\mathbf{r}_P(0) = 5\mathbf{i} + 6\mathbf{j}$. 2. **Condition for interception:** For interception to occur at time t, the displacement of the patrol boat from its starting position must equal the relative position vector.

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43 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Six-digit numbers are formed using digits chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$, with no digit repeated in any number. Find the number of different six-digit odd numbers that can be formed such that the digits 2 and 4 are not adjacent.

a. 8160 ✓

b. 7680

c. 9120

d. 2880

We use complementary counting: $\text{Number of valid arrangements} = \text{Total odd 6 – digit numbers} - \text{Odd 6 – digit numbers where 2 and 4 are adjacent}$. **Step 1: Total number of 6-digit odd numbers** - For the number to be odd, the last digit (units place) must be chosen from the odd digits $\{1, 3, 5, 7\}$, giving 4 choices. - The remaining 5 positions are filled by selecting and arranging 5 digits from the remaining 7 digits in the set.

44 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

The real numbers x and y satisfy the simultaneous equations: $\log_3(x) + 2\log_9(y-1) = 2$ $4^y \cdot 8^{-\frac{x}{3}} = 32$ Find the value of $x^2 + y^2$.

a. 25 ✓

b. 10

c. $\frac{145}{4}$

d. 50

Step 1: Simplify the logarithmic equation using the change-of-base formula. Recall that $\log_9(y-1) = \frac{\log_3(y-1)}{\log_3(9)} = \frac{\log_3(y-1)}{2}$. Substituting this gives: $\log_3(x) + 2\left(\frac{\log_3(y-1)}{2}\right) = 2 \Rightarrow \log_3(x) + \log_3(y-1) = 2$ Applying the product rule of logarithms: $\log_3[x(y-1)] = 2 \Rightarrow x(y-1) = 3^2 = 9 \quad - - - \text{(Equation 1)}$ Step 2: Simplify the exponential equation using laws of indices. Express all terms wi.

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45 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

A supply launch S departs from a base at the origin O at time $t = 0$ to intercept an automated cargo barge C. At $t = 0$, the barge has position vector $\mathbf{r}_0 = (16; 12)$ km relative to O and maintains a constant velocity of $\mathbf{v}_C = (-15; 0)$ km/h. The launch travels in a straight line with constant velocity $\mathbf{v}_S$ and constant speed v km/h. Determine the minimum speed v required for the launch to successfully intercept the barge.

a. 9 km/h ✓

b. 12 km/h

c. 15 km/h

d. 7.2 km/h

To intercept the cargo barge at some time $t > 0$, the position vectors must satisfy: $\mathbf{r}_S(t) = \mathbf{r}_C(t) \Rightarrow t\mathbf{v}_S = \mathbf{r}_0 + t\mathbf{v}_C$ Dividing by t gives: $\mathbf{v}_S = \mathbf{v}_C + \frac{1}{t}\mathbf{r}_0 = (-15; 0) + \frac{1}{t}(16; 12) = \left(-15 + \frac{16}{t}; [4pt]\frac{12}{t}\right)$ Let $u = \frac{1}{t} > 0$. The square of the speed of the launch is.

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